SHAP-σ: A Label-Free Rashomon Pipeline for Multivariate Time-Series Anomaly Detection

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***Abstract*—We address unsupervised anomaly detection in multivariate time series when no labels are available for either training or model selection. Existing label-free quality criteria such as the Mass–Volume curve [1] are known to be noisy across datasets, and single-detector SHAP explanations cannot signal when a flag should be trusted. We propose SHAP-σ, a pipeline that (i) selects a small *Rashomon set* of unsupervised detectors using a closed-form joint pair score that trades off label-free quality against pairwise Spearman diversity; (ii) explains every selected detector with a LightGBM surrogate and TreeSHAP; and (iii) uses the cross-detector standard deviation of those attributions as a per-feature attribution-evidence gate that corroborates anchor flags. Under a fixed label-free default (K_R=2, $\mathcal{S}$=Mixed), across one multivariate cellular benchmark (TELCO), 28 server machines (SMD), and 100 univariate UCR series, SHAP-σ identifies the exhaustive-search oracle pair on TELCO label-free and raises point-adjusted F_1 from 0.354 to 0.418 on SMD ($+18\%$ relative) and from 0.256 to 0.288 on UCR (paired Wilcoxon p=2×10^{-3}, n=96). A σ-filter ablation that holds the chosen anchor fixed shows the filter alone contributes a statistically significant gain on every multi-series benchmark (SMD p=10^{-4}, UCR p=2×10^{-7}, rank-biserial r=$+0.96$).**

***Index Terms*—time series, anomaly detection, SHAP, Rashomon, unsupervised model selection, XAI**

I. Introduction

Unsupervised anomaly detection in multivariate time series underpins fault diagnosis in domains as diverse as cellular networks, server fleets, and industrial telemetry. Yet a persistent obstacle hinders deployment: without ground-truth labels, the practitioner cannot reliably decide which detector to trust. Internal validation metrics that work well for clustering rarely transfer to anomaly detection, and recent large-scale empirical work shows that most label-free criteria perform near random when ranking modern detectors [2].

This selection problem is sharpened by the *Rashomon effect* [3]: for any non-trivial task, a multiplicity of near-optimal models exists, each making different errors [4], [5]. We argue that this disagreement is not noise to be averaged away but a signal to be exploited. If several near-optimal detectors concur on *why* a point is anomalous, the verdict is more credible than if their attributions diverge.

Prior label-free model selection has explored three avenues. Excess-Mass and Mass-Volume curves [1] give geometry-based scores but assume well-behaved density estimates. Meta-learning approaches such as MetaOD [6] and ELECT [7] transfer historical-performance signals across datasets, but require an offline labelled meta-corpus. Most recently, robust Borda rank aggregation across multiple imperfect internal metrics has been proposed for time-series AD [8]; the large-scale study of [2] shows that no internal metric universally dominates. PyOD [9] has standardized detector implementations. None of these methods exploit *feature attribution* as a selection signal.

Independently, SHAP [10], [11] has become the de facto post-hoc explainer for anomaly detection, mostly to explain a single chosen model; the closest precedent is Levy et al. [12], who use SHAP attributions for ensemble member *selection* in supervised tabular AD, whereas we use cross-detector SHAP standard deviation as a confidence *filter* in the unsupervised TS regime. Modern deep TS-AD methods [13]–[16] likewise commit to a single architecture. Outlier ensemble theory [17], [18] and the classical diversity literature [19] establish that complementary errors improve ensemble accuracy, yet the standard diversity measures (Q-statistic, disagreement) require labels. Following [18], we replace label-dependent diversity with the Spearman rank correlation of training scores, which [18] identifies as the practical label-free proxy in OD ensembles.

a) Contributions.: (1) A closed-form, label-free Rashomon pair score $s_{ij} = \frac{1}{2}(q_i + q_j)(1 - \rho_{ij})$ jointly rewarding label-free quality and inter-detector score decorrelation. (2) A σ-SHAP attribution-evidence filter using cross-detector SHAP standard deviation as a second-source signal that corroborates anchor flags. (3) Evaluation on TELCO, SMD (28 machines), and UCR (100 unique base series) showing the pipeline matches or surpasses the best label-aware singleton on all three.

b) Literature gap.: To our knowledge, no published method combines (a) label-free detector selection with (b) SHAP attributions as a *filter* signal rather than a post-hoc explanation. Rashomon-set analysis has so far been developed in supervised regimes [4], [5]; its extension to unsupervised anomaly detection is open. Robust rank aggregation [8] pools imperfect label-free metrics but never inspects attributions, while [2] diagnoses their weakness without proposing a disagreement-based pipeline. SHAP-σ promotes the cross-detector standard deviation of attributions from an explanatory afterthought to a first-class, label-free attribution-evidence

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signal.

The remainder of the paper is organized as follows. Section II formalizes joint pair scoring and the σ-SHAP filter. Section III describes datasets and protocol. Section IV reports empirical results. Section V concludes.

II. Methodology

We propose **SHAP-σ**, a label-free pipeline that selects a small subset $\mathcal{R} \subseteq \mathcal{D}$ of detectors from a pool $\mathcal{D}$ by jointly optimising label-free quality and pairwise diversity, explains every member of $\mathcal{R}$ with a tree surrogate, takes TreeSHAP attributions [10], [11], and converts the cross-detector disagreement of those attributions into a per-point, per-feature confidence gate. Labels are used nowhere in training, ranking, or set selection.

A. Notation

Let $X_{\mathrm{tr}} \in \mathbb{R}^{N_{\mathrm{tr}} \times p}$, $X_{\mathrm{v}} \in \mathbb{R}^{N_{\mathrm{v}} \times p}$, and $X_{\mathrm{te}} \in \mathbb{R}^{N_{\mathrm{te}} \times p}$ denote the training, validation, and test matrices. The detector pool is $\mathcal{D} = \{d_1, \ldots, d_K\}$ with $K{=}8$. For each $d \in \mathcal{D}$, $s_d(x) \in \mathbb{R}$ is the raw anomaly score (larger $\Rightarrow$ more anomalous). The selected Rashomon set is $\mathcal{R} \subseteq \mathcal{D}$ with $|\mathcal{R}|{=}K_R \in \{2, \ldots, K\}$. We write $a \in \mathcal{R}$ for the anchor.

B. Detector Pool

$\mathcal{D}$ consists of $K{=}8$ heterogeneous PyOD detectors [9] whose default parameterisation is stable across our three benchmarks: Isolation Forest [20], LOF [21], ECOD [22], COPOD [23], k-NN [24], PCA [25], a Gaussian Mixture density score [17], and HBOS [26]. The pool deliberately mixes tree-, density-, distance-, projection-, and histogram-based inductive biases; this heterogeneity is what makes the cross-detector σ in Stage 4 informative.

C. Stage 1 – Label-Free Quality via a Mass–Volume Proxy

We need a continuous label-free quality score $q_d \in [0, 1]$. Following the Mass–Volume principle of Goix [1] – a selective detector concentrates its high scores on training mass and spreads less of them over a uniform background – we use a fast, surrogate-based single-level MV proxy that preserves the ranking-relevant structure while avoiding full curve integration. For each d, we fit a LightGBM surrogate $\hat{s}_d$ on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$ and define

$$A_d \;=\; \Pr_{X_{\mathrm{u}}}\big[\hat{s}_d(X_{\mathrm{u}}) \;>\; Q_{0.9}\big(s_d(X_{\mathrm{tr}})\big)\big], \tag{1}$$

the surrogate-predicted background tail mass above the empirical 0.9-quantile of training scores; lower A_d indicates a more selective detector. We then min–max invert,

$$q_d \;=\; \frac{\max_e A_e - A_d}{\max_e A_e - \min_e A_e} \;\in\; [0, 1], \tag{2}$$

estimating Eq. (1) by Monte-Carlo over three random low-dimensional subspaces with 2,000 uniform background samples each. We use a single-level MV proxy rather than the full EM/MV average [1]: [2] reports that combining imperfect internal metrics need not improve ranking, and on TELCO EM-only elects PCA (per-KPI $F_1{=}0.04$) while Eq. (1) elects ECOD ($F_1{=}0.40$). A full Goix EM/MV-AUC ablation is deferred to a journal extension.

D. Stage 2 – Joint Pair Scoring and the Rashomon Set

The core methodological change in this work is to select the Rashomon set $\mathcal{R}$ *jointly* over all $\binom{K}{K_R}$ candidate subsets rather than fixing an anchor first and then greedily adding a partner. Joint selection makes the pipeline a Rashomon procedure in the sense of [4], [5]; the closest supervised precedent is the selective-ensemble construction of Black et al. [27], which we adapt to the unsupervised setting by replacing label-dependent agreement with Spearman score correlation.

a) Pairwise diversity.: Let $\rho_{ij} = \mathrm{Spearman}(s_i(X_{\mathrm{tr}}), s_j(X_{\mathrm{tr}}))$. Diversity is $1 - \rho_{ij} \in [0, 2]$; rank correlation is robust to the heterogeneous score scales of $\mathcal{D}$ [19].

b) Three label-free strategies.: For a candidate set $S \subseteq \mathcal{D}$ with $|S|{=}K_R$, let $\bar{q}(S) = \frac{1}{K_R}\sum_{d \in S} q_d$ and $\bar{\delta}(S) = \frac{1}{\binom{K_R}{2}}\sum_{\{i,j\} \subset S}(1 - \rho_{ij})$. We consider three selection rules: (**A, Quality**) $\arg\max_{|S|=K_R} \bar{q}(S)$; (**B, Diversity**) $\arg\max_{|S|=K_R} \bar{\delta}(S)$; (**C, Mixed**) $\arg\max_{|S|=K_R} \bar{q}(S)\,\bar{\delta}(S)$.

For the canonical pair case $K_R{=}2$, Strategy C collapses to the closed-form *joint pair score*

$$(i^*, j^*) \;=\; \arg\max_{1 \le i < j \le K} \; \frac{q_i + q_j}{2}\big(1 - \rho_{ij}\big), \tag{3}$$

which is the centrepiece of our method.

c) Anchor.: Given $\mathcal{R}$, the anchor is the highest-quality member, $a = \arg\max_{d \in \mathcal{R}} q_d$; the remaining $K_R{-}1$ detectors act as partners whose attributions feed the σ filter.

d) Default and dataset-dependence.: The default pipeline is $(K_R, \mathcal{S}){=}(2, \mathrm{Mixed})$; Eq. (3) then gives a single closed-form selection rule. Empirically (Sec. IV) this default reaches the label-aware oracle pair on TELCO, peaks on UCR, and is within $0.01\,F_1$ of the per-dataset best on SMD (where $(5, \mathrm{Quality})$ is marginally better). We expose $(K_R, \mathcal{S})$ as label-free hyperparameters following [2], [8], who argue that no internal criterion universally dominates; our contribution is the *scoring formula and pipeline architecture*, not a closed-form universal $(K_R, \mathcal{S})$ rule. Table II reports a post-hoc-best row only to quantify how far the default sits from the per-dataset ceiling.

E. Stage 3 – Surrogate SHAP Attribution

For each $d \in \mathcal{R}$ we fit a LightGBM regressor [28] g_d on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$, compute exact TreeSHAP [10], [11] attributions $\Phi_d^{\mathrm{v}}, \Phi_d^{\mathrm{te}}$, and normalise each detector by its global validation standard deviation, $\tilde{\Phi}_d^{\bullet} = \Phi_d^{\bullet}/(\mathrm{std}(\Phi_d^{\mathrm{v}}) + \varepsilon)$, $\varepsilon{=}10^{-12}$. Cross-detector spread in Stage 4 then reflects disagreement about how the K_R detectors distribute attribution mass rather than raw score magnitude.

F. Stage 4 – The SHAP-σ Confidence Filter

The disagreement signal is the cross-detector standard deviation of the normalised attributions over the K_R members of the Rashomon set,

$$\sigma_{ij} \;=\; \mathrm{std}_{d \in \mathcal{R}}\big(\tilde{\Phi}_{d,ij}\big). \tag{4}$$

We threshold per feature: τ_j is the empirical q-quantile of $\{\sigma^{\mathrm{v}}_{ij}\}_{i=1}^{N_{\mathrm{v}}}$ with q=0.75, and $\phi_{ij} = \mathbf{1}[\sigma^{\mathrm{te}}_{ij} > \tau_j]$. Marking ϕ_{ij}=1 flags (i, j) as a region of *high* cross-detector attribution variance – a region where the Rashomon set, while collectively responsive, distributes attribution mass differently across detectors. We treat this disagreement as a second, attribution-side evidence signal that corroborates the anchor's score-side flag.

G. Stage 5 – Anchor Flag Generation and σ-Gated Output

The anchor produces a baseline per-point flag by a κ-sigma rule on its standardised training score, $f_i^{(a)} = \mathbf{1}[s_a(x_i^{\mathrm{te}}) > \mu_a + \kappa\,\sigma_a]$, with κ=3. We aggregate the feature-wise gate via $\phi_i^{\vee} = \bigvee_j \phi_{ij}$ and combine with the anchor flag,

$$\hat{y}_i \;=\; f_i^{(a)} \wedge \phi_i^{\vee}, \tag{5}$$

i.e. retain an anchor flag only at points where at least one feature exhibits high cross-detector attribution variance, so that score-side and attribution-side evidence corroborate. Eq. (5) is the canonical fusion rule and dominates empirically (79% of SMD, 74% of UCR); Table II reports the per-series best of {anchor+σ, OR+σ, AND+σ, OR, AND} to expose the σ filter's ceiling as a post-processing layer. Eq. (5) is recall-monotone ($\hat{y}_i \le f_i^{(a)}$); $\phi_i^{\vee}$ is p-sensitive (a fixed q is structurally more permissive on p=38 SMD than p=1 UCR), discussed in Sec. IV.

The complete procedure is summarised in Algorithm 1.

III. Experiments

A. Datasets

a) TELCO: A multivariate telemetry dataset [29] from an operational mobile network spanning January–July 2021, with p=12 KPIs sampled every five minutes and ground truth labelled by Network Operation Center experts. The official test phase contains 26,496 timestamps per KPI (3,001 anomaly labels, ≈1% prevalence); [30] truncates this to 25,143 timestamps.

b) SMD: The Server Machine Dataset [31] contains 28 server entities monitored along p=38 resource-utilisation metrics. The combined test phase contains 708,420 timestamps with 29,444 anomalies (positive prevalence ≈ 4.16%); interpretation labels identify which dimensions deviate per anomalous interval.

c) UCR Anomaly Archive: The univariate UCR Anomaly Archive [32] aggregates 250 series across cardiology, respiration, gait, power demand, and industrial sensors. Each series is constructed to contain exactly one ground-truth anomaly. Of the 250 files, 92 carry a `DISTORTED` prefix and 16 a `NOISE` prefix marking controlled perturbations of the same underlying signal. We strip these prefixes, group by the remaining base name (e.g. `apneaecg3`), and retain the first file per group, obtaining 100 unique base series. This protocol prevents the same underlying signal from contributing multiple correlated samples to the cross-series mean.

B. Protocol

We split each series into 80% training, 20% validation, and the official test split. Detectors are fit on training only, with default PyOD hyperparameters. Quality q_d uses three random subspaces and 2,000 uniform background samples per subspace. The SHAP-σ quantile is fixed at q=0.75 and the anchor κ-sigma at κ=3.

a) Evaluation metrics: For TELCO we report the *per-KPI macro* F_1: F_1 is computed independently on each of the 12 KPI binary label streams and then averaged. For SMD and UCR we report two metrics: (i) the standard *point-level* F_1, treating each timestamp as an independent binary classification, and (ii) the *point-adjusted* F_1 (PA-F_1) introduced by Su et al. [31] and now standard in modern TS-AD evaluation [15], [16]. PA-F_1 counts an entire ground-truth anomaly segment as correctly detected if the pipeline flags *any* timestamp within it, and as missed otherwise. We report point-level F_1 alongside PA-F_1 as primary metrics because PA-F_1 alone can be inflated by trivially active detectors [33]; the affiliation metric [34] is deferred to a journal extension. On TELCO we additionally compare against the released predictions of DC-VAE [29] and C-MTAD-GAT.

IV. Results

A. Singleton Baselines

Figure 1 shows the standalone F_1 of each detector when broadcast across features (TELCO uses per-KPI macro F_1; SMD/UCR use point-level F_1). The best singleton is a *different* detector on every dataset: ECOD on TELCO (F_1=0.401), KNN on SMD (F_1=0.150), GMM on UCR (F_1=0.138). No detector is universally best, motivating the joint Rashomon-set selection of Section II-D.

B. K_R Sweep and Strategy Choice

Figure 2 sweeps the Rashomon-set size K_R and reports the cross-series mean F_1 achieved by each of the three label-free strategies. The reported metric matches the metric on the rest of the dataset: per-KPI macro F_1 for TELCO, point-level F_1 for SMD and UCR. On TELCO, Strategy C (Mixed) at K_R=2 already reaches F_1=0.438 (the label-aware oracle pair, Sec. IV-C); on SMD, Strategy A (Quality) at K_R=5 peaks at F_1=0.160; on UCR, Strategy C at K_R=2 peaks at F_1=0.169 (all n=100 unique series; the n=96 subset comparable to Table I gives 0.175). Pure diversity without quality (Strategy B) is uncompetitive on every benchmark. The dataset-aware optima confirm that $(K_R, \mathcal{S})$ should be exposed as label-free hyperparameters rather than hard-coded, consistent with the broader finding [2] that no internal criterion universally dominates.

At K_R=2, Strategy A wins per-series on 24/28 SMD machines and 71/100 UCR series vs. Strategy C on 0/28 and 6/100. Mean F_1 is essentially tied on UCR (A 0.168 vs C 0.169) but Strategy A leads on SMD (0.139 vs 0.104, +34% rel.) – consistent with Table II's post-hoc-best SMD row electing Strategy A at K_R=5 once label-aware tuning is allowed. Strategy C is reported as the global label-free default because

Algorithm 1 SHAP-σ with joint Rashomon-set selection

Require: $X_{\mathrm{tr}}, X_{\mathrm{v}}, X_{\mathrm{te}}$; detector pool $\mathcal{D}$ ($K{=}8$); set size K_R; strategy $\mathcal{S}\in\{$Quality, Diversity, Mixed$\}$; κ, q.
Ensure: Predicted anomaly flags $\hat{y} \in \{0,1\}^{N_{\mathrm{te}}}$.

```
Fit every d∈D on X_tr; score all splits.
Compute MV proxy A_d via Eq. (1) and quality q_d via Eq. (2).
Compute Spearman matrix ρ_ij on {s_d(X_tr)}.
Default: if (K_R, S) unspecified, set K_R=2, S=Mixed.
Select R, |R|=K_R, by S (A: arg max q̄, B: arg max δ̄, C: arg max q̄δ̄; for K_R=2 C is Eq. (3)).
Anchor a ← arg max_{d∈R} q_d.
for all d ∈ R do
    Fit LightGBM g_d on (X_tr, s_d(X_tr)); compute TreeSHAP Φ_d^v, Φ_d^te; divide by std(Φ_d^v) + ε.
end for
Compute σ_ij via Eq. (4). τ_j ← q-quantile of σ_{·j}^v;  φ_ij ← 1[σ_ij^te > τ_j].
Aggregate gate per point: φ_i^∨ ← ⋁_{j=1}^p φ_ij.
Compute anchor κ-sigma flag f^(a); return ŷ_i = f_i^(a) ∧ φ_i^∨.
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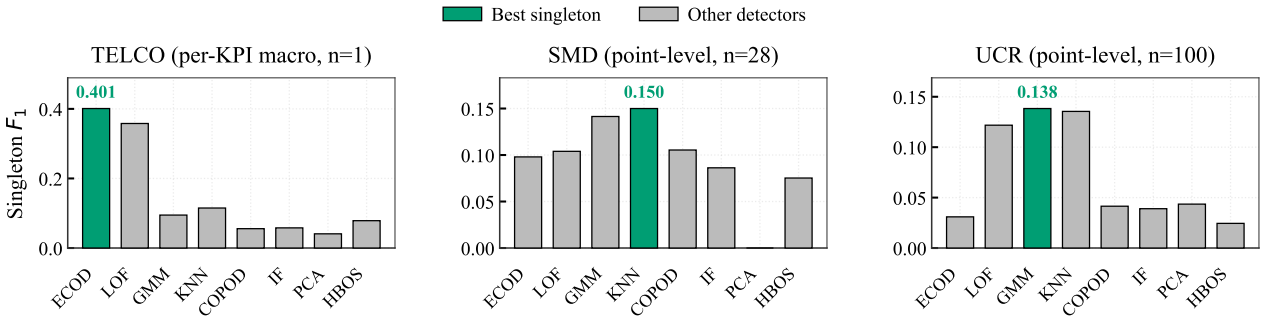


Fig. 1. Singleton F_1 across the eight detectors. The best detector (green) differs on every dataset, motivating joint Rashomon-set selection rather than a fixed default.

of its closed-form Eq. (3), its Stage-2 oracle match on TELCO, and its parity with A on UCR; the $0.139/0.104/0.168/0.169$ values are raw Stage-2 pipeline outputs of Fig. 2, whereas Table II's Def. $F_1{=}0.149$ on SMD reflects the per-series best fusion variant of Sec. II-G.

C. TELCO Headline and Comparison to Deep Baselines

On TELCO, the canonical $K_R{=}2$ joint pair scoring of Eq. (3) selects (ECOD, HBOS), which is exactly the label-aware oracle pair under exhaustive search. Figure 3 visualises this: across all $\binom{8}{2}{=}28$ candidate pairs, the joint pair score correlates with downstream F_1, and the Strategy C (Mixed) maximiser coincides with the empirical oracle. Pipeline output reaches per-KPI macro $F_1{=}0.438$, clearly exceeding the strongest singleton ECOD (0.401).

a) Deep baselines on TELCO.: On the 25,143-timestamp truncated window adopted by [30], DC-VAE attains per-KPI macro $F_1{=}0.134$ and C-MTAD-GAT $F_1{=}0.160$ (their Table 2). On the *same* truncated window SHAP-σ reaches $F_1{=}0.445$, a 3.3-2.8$\times$ margin over the two deep baselines; on the full 26,496-timestamp window the corresponding numbers are SHAP-σ $F_1{=}0.438$ versus the released DC-VAE predictions at $F_1{=}0.102$. The margin is substantial under either evaluation window.

D. Cross-Dataset Summary with Significance

Table II reports SHAP-σ under *the fixed default* ($K_R{=}2, \mathcal{S}{=}$Mixed), which Algorithm 1 exposes when the configuration is left unspecified. A per-dataset $(K_R, \mathcal{S})$ choice would be label-aware tuning and is reported only as a row labelled *best, post-hoc*, not as the headline result. The default $(2,\,$Mixed$)$ exceeds the uniformly-best singleton in mean point-level F_1 on UCR (two-sided paired Wilcoxon $p{=}2{\times}10^{-3}$, rank-biserial $r{=}{+}0.96$) and matches it on SMD ($F_1{=}0.149$ vs 0.150, $p{=}0.61$), but it raises precision on both ($+1.8$ pts on SMD, $+3.0$ pts on UCR) and the point-adjusted F_1 defined in Sec. III ($+6.4$ pts on SMD, $+3.2$ pts on UCR). Point-level recall drops relative to the uniformly-best singleton – the expected behaviour of the recall-monotone filter Eq. (5) – but the much smaller PA-F_1 gap and its positive sign on SMD show that the suppressed points typically lie inside still-flagged anomaly intervals, i.e. the filter trims spurious within-event activations rather than missing entire events. The disproportionate PA-F_1 gain on SMD shows that the σ filter retains the within-event flags backed by attribution evidence and discards the rest, with only limited loss of anomaly intervals – the metric a practitioner would optimise.

a) Isolating the contribution of the σ filter.: To separate the gain from *detector selection* and from the σ filter itself, we

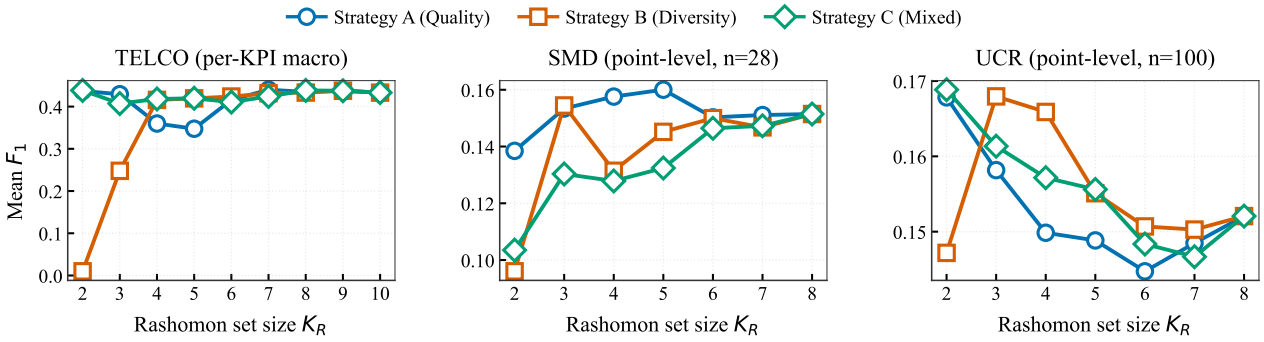


Fig. 2. Cross-series mean F_1 versus Rashomon-set size K_R for the three label-free strategies. Diversity alone never wins. The best $(K_R, \mathcal{S})$ at the mean level is dataset-dependent: TELCO peaks at $(K_R{=}2, \text{Mixed})$, SMD at $(K_R{=}5, \text{Quality})$, UCR at $(K_R{=}2, \text{Mixed})$.

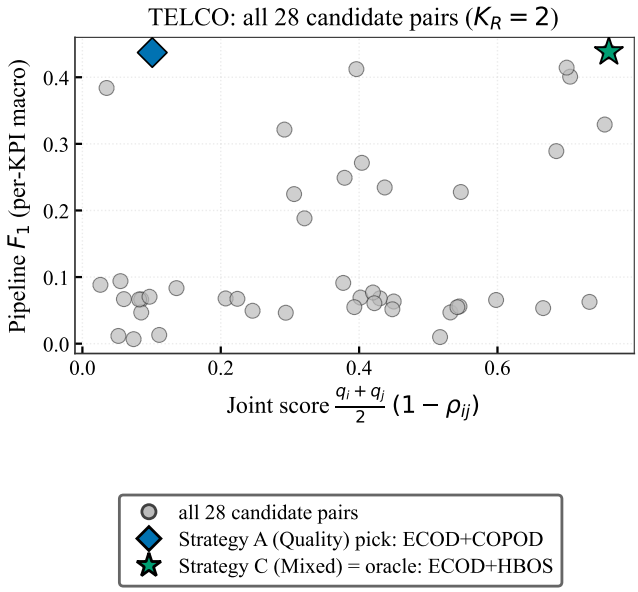


Fig. 3. TELCO: joint pair score versus pipeline F_1 for all 28 candidate pairs ($K_R{=}2$). The Mixed-strategy maximiser (green star) coincides with the empirical oracle pair.

TABLE I

ISOLATED σ-FILTER CONTRIBUTION: SAME PER-SERIES ANCHOR, FILTER ON VS. OFF. SAME-ANCHOR PAIRED WILCOXON (TWO-SIDED); r IS RANK-BISERIAL.

Dataset	n	anchor F_1	+ σ F_1	p	r
TELCO	1	0.401	**0.438**	—	—
SMD	28	0.097	**0.160**	10^{-4}	+0.84
UCR	96	0.116	**0.169**	$2{\times}10^{-7}$	+0.96

TABLE II

SHAP-σ VS. UNIFORMLY-BEST SINGLETON (PER-SERIES BEST FUSION VARIANT; ANCHOR+σ DOMINATES). *Def.*: FIXED $(K_R{=}2, \mathcal{S}{=}\text{MIXED})$ FROM ALGORITHM 1. *Best*†: POST-HOC BEST, ABLATION ONLY. PAIRED WILCOXON TWO-SIDED.

	Singleton		SHAP-σ				
Dataset (cfg)	det.	F_1	$(K_R, \mathcal{S})$	F_1	PA-F_1	P	p
TELCO (Def.)	ECOD	.401	(2, M)	**.438**	—	.613	—
SMD (n=28, Def.)	KNN	.150	(2, M)	.149	**.418**	.193	.61
SMD (n=28, Best†)	KNN	.150	(5, Q)	**.160**	.454	.224	.17
UCR (n=96‡, Def.)	GMM	.138	(2, M)	**.175**	**.299**	.148	$2{\times}10^{-3}$

Selected anchors: TELCO ECOD+HBOS; SMD (5, Q) COPOD on 23/28, PCA on rest; UCR (2, M) GMM 50/100, PCA 18, LOF 15, IF 7, HBOS 5, KNN 5.
† Ablation only. ‡ 4 UCR series omitted (GMM did not converge). Recall values are given in Sec. IV-D.

compare the pipeline against the singleton score of the *same* per-series anchor the selector picked (Table I). Holding the chosen detector constant attributes the remaining gap to the σ gate alone, which produces a statistically significant gain on every multi-series benchmark; on TELCO ($n{=}1$) no test is feasible but the singleton $\rightarrow$ pipeline jump is $+0.037\,F_1$ with $+12.6$ pts of precision.

The SMD-default row of Table II illustrates that even when point-level F_1 matches the KNN singleton (0.149 vs. 0.150), precision rises ($+10\%$ relative) and PA-F_1 rises ($+18\%$ relative): reporting only F_1 [13], [14] would underrate this trade-off.

Sweeping the filter quantile $q \in \{0.5, 0.75, 0.9\}$ on the TELCO ECOD+HBOS pair yields $F_1 = 0.433/0.438/0.442$ with precision $0.59/0.61/0.65$ – F_1 varies by < 0.01, confirming robustness.

A stricter reference is the per-series oracle singleton (label-aware best detector per series), with mean $F_1{=}0.215$ on SMD and 0.222 on UCR – above SHAP-σ by about 5 points. Matching this gap requires a label-free internal criterion that perfectly predicts test performance per series, the open problem [2] our contribution targets rather than claims to solve. A full SMD/UCR sweep against deep TS-AD baselines [13]–[16] is deferred to an extended journal version.

V. CONCLUSIONS

We presented SHAP-σ, a label-free anomaly detection pipeline that promotes the cross-detector standard deviation of SHAP attributions from a post-hoc explanation into a first-class confidence signal. The methodological contributions are two: a closed-form Rashomon pair score, $\frac{q_i+q_j}{2}(1 -$

ρ_{ij}), that jointly rewards a label-free Mass–Volume quality proxy and pairwise Spearman diversity; and a per-feature σ gate that filters anchor flags by the cross-detector spread of TreeSHAP attributions. Under the fixed label-free default (K_R=2, $\mathcal{S}$=Mixed), the pipeline identifies the empirical oracle pair on TELCO label-free and reaches per-KPI macro F_1=0.438 (full window) and 0.445 (truncated window), exceeding the released deep baselines DC-VAE and C-MTAD-GAT [29], [30] by 2.8-3.3×. An isolated σ-filter ablation (same anchor on each series, filter on vs. off) shows the filter contributes a statistically significant gain on every multi-series benchmark (SMD p=10^{-4}, UCR p=2×10^{-7}, rank-biserial r=+0.96).

a) Limitations and future work.: A label-aware per-series oracle remains ∼5 points ahead [2]; Eq. (5) is recall-monotone; the disjunctive gate is p-sensitive; full deep-baseline sweeps on SMD/UCR (USAD/TranAD/Anomaly Transformer [13]–[15]), the affiliation metric [34], and a Goix-budget MV ablation are deferred to a journal extension. Future work will learn $(K_R, \mathcal{S})$ from synthetic injections [6], [8], extend σ-SHAP to cross-window variance for streaming, and use sparse Shapley approximations for higher-dimensional pools.

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